Objective of the LAB.

To deploy “http” service on 2 EC2 instance from CodeDeploy and also copy an test page to the EC2 instance.

Result:

We should have an web service running on the EC2 and we should be able to access it via the EC2 instance public IP.

Lab-Setups – CodeDeploy

1. Create IAM Policy & Role for EC2 to access S3

    1. Provide – S3 – get and list Permission for the EC2 instance. -- EC2-codedeploy

    Example for the name , You can have same name for Role and Policy,eg: -- EC2-codedeploy

2. Create Ec2 instance with “EC2-codedeploy” Roles attached to it and install the Codedeploy agent along with it (Bootstrap option)

3. Upload the “Sample-linuz.zip” into S3 bucket and give “Public” access

4. Create IAM Policy & Role for CodeDeploy to Access EC2 instance

    Create Roles ,eg: -- and CodeDeploy

5. Create an CodeDeploy deployment code.

6. Create an ‘New Deploy” and point to the S3 location of the Sample-linux.zip file

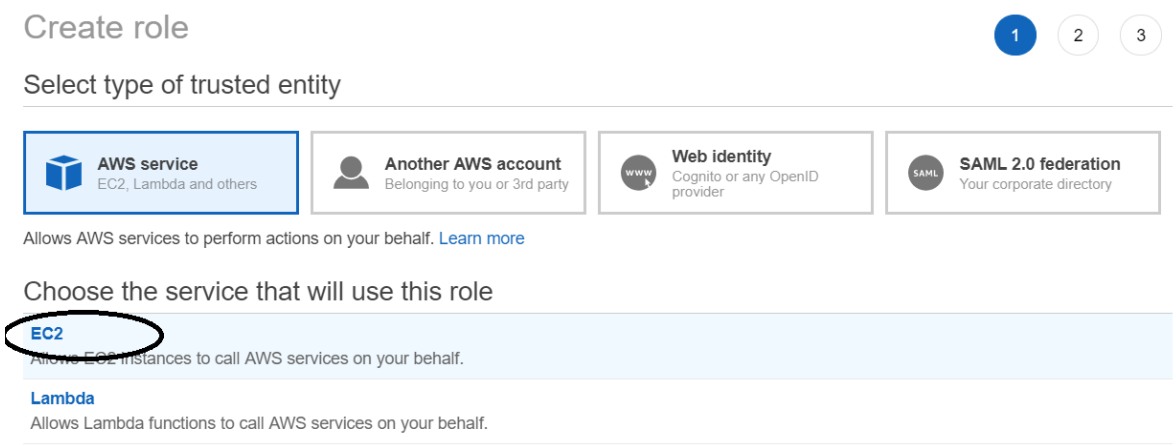
7. Update the index.html file in the “.zip” file, and change “Congratulations” Edit the “appsepc” file. Remove all the lines from “hooks” and save it.

Step 1. Policy Creation

a. **IAM Policy** -- **EC2-codedeploy** for EC2 to access S3

```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Action": [
                "s3:Get*",
                "s3:List*"
            ],
            "Effect": "Allow",
            "Resource": "*"
        }
    ]
}
```

b. IAM Role for **EC2-codedeploy**

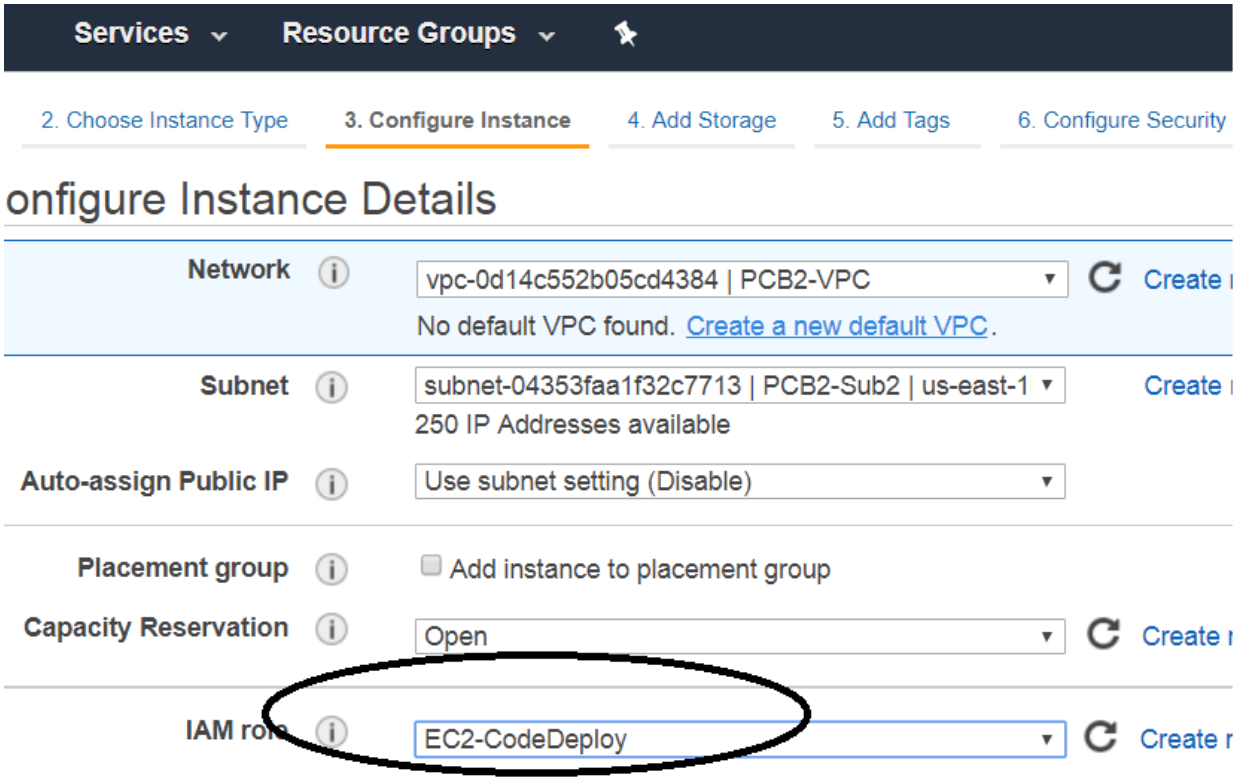


Then attach the policy "**EC2-codedeploy**"

Step 2. Create EC2 instance

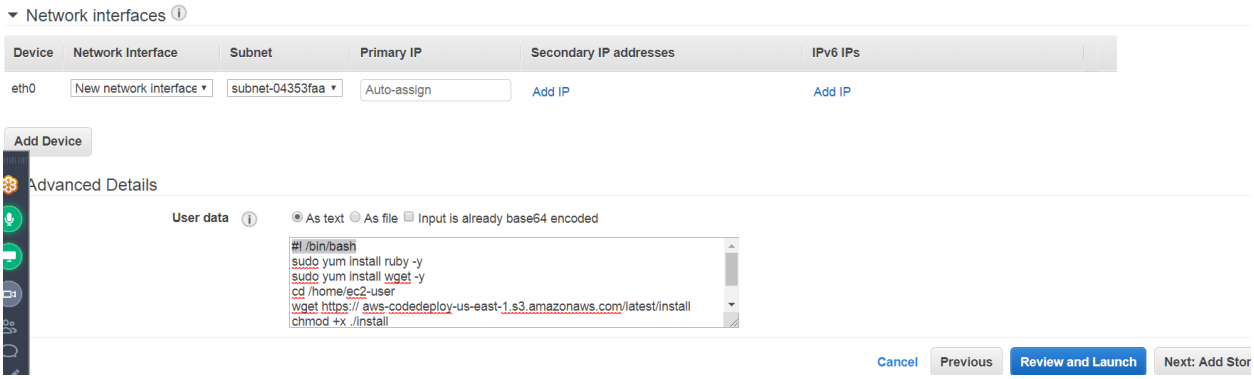
Create 2 Instance with the Above IAM role that was created for EC2 "**EC2-codedeploy**"

I.



II.

Under the Advance option copy the Below command along with it



#!/bin/bash

sudo yum install ruby -y

sudo yum install wget -y

cd /home/ec2-user

wget https://**aws-codedeploy-us-east-1**.s3.amazonaws.com/latest/install

chmod +x ./install

sudo ./install auto

This would automatically install the Codedeploy agent and its called as “Bootstrapping”

III. With Security Group allowing -- SSH and HTTP traffic

IV. After the EC2 instance is up, check if the codedeploy agent service is running with the below command.

 # service codedeploy-agent status.

Output:

```
[root@ip-172-31-83-93 ec2-user]# service codedeploy-agent status
The AWS CodeDeploy agent is running as PID 2704
[root@ip-172-31-83-93 ec2-user]#
```

Resource Kit Bucket Names by Region

This table lists the names of `bucket-name` replacements required for some procedures in the guide. These are the names of the Amazon S3 buckets that contain the AWS CodeDeploy Resource Kit files.

Region name	`bucket-name` replacement	Region identifier
US East (Ohio)	aws-codedeploy-us-east-2	us-east-2
US East (N. Virginia)	aws-codedeploy-us-east-1	us-east-1
US West (N. California)	aws-codedeploy-us-west-1	us-west-1
US West (Oregon)	aws-codedeploy-us-west-2	us-west-2
Canada (Central)	aws-codedeploy-ca-central-1	ca-central-1
EU (Ireland)	aws-codedeploy-eu-west-1	eu-west-1
EU (London)	aws-codedeploy-eu-west-2	eu-west-2
EU (Paris)	aws-codedeploy-eu-west-3	eu-west-3
EU (Frankfurt)	aws-codedeploy-eu-central-1	eu-central-1
Asia Pacific (Tokyo)	aws-codedeploy-ap-northeast-1	ap-northeast-1
Asia Pacific (Seoul)	aws-codedeploy-ap-northeast-2	ap-northeast-2
Asia Pacific (Singapore)	aws-codedeploy-ap-southeast-1	ap-southeast-1
Asia Pacific (Sydney)	aws-codedeploy-ap-southeast-2	ap-southeast-2
Asia Pacific (Mumbai)	aws-codedeploy-ap-south-1	ap-south-1
South America (São Paulo)	aws-codedeploy-sa-east-1	sa-east-1

Step 3.

Upload the “Sample-linuz.zip” into S3 bucket and give “Public” access.

The bucket needs to have "Versioning" ENABLED.

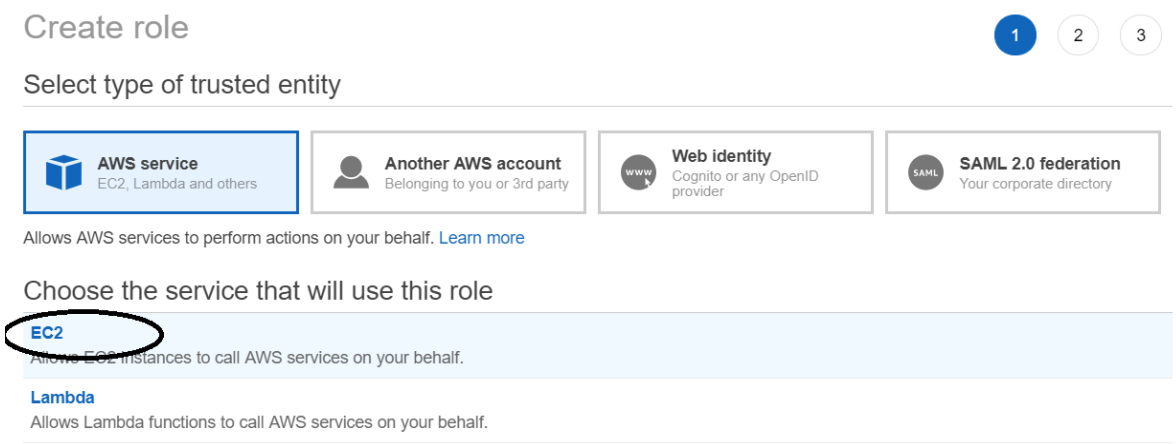
The file is available in the Google drive
https://drive.google.com/drive/u/1/folders/18wNY2GTIcS3vCufomVo9If5B59P5l8nS

Step 4: Create IAM Role and Policy for **CodeDeploy**

a. IAM Policy -- **Codedeploy** to access EC2 instance.
Eg:-- "Codedeploy"

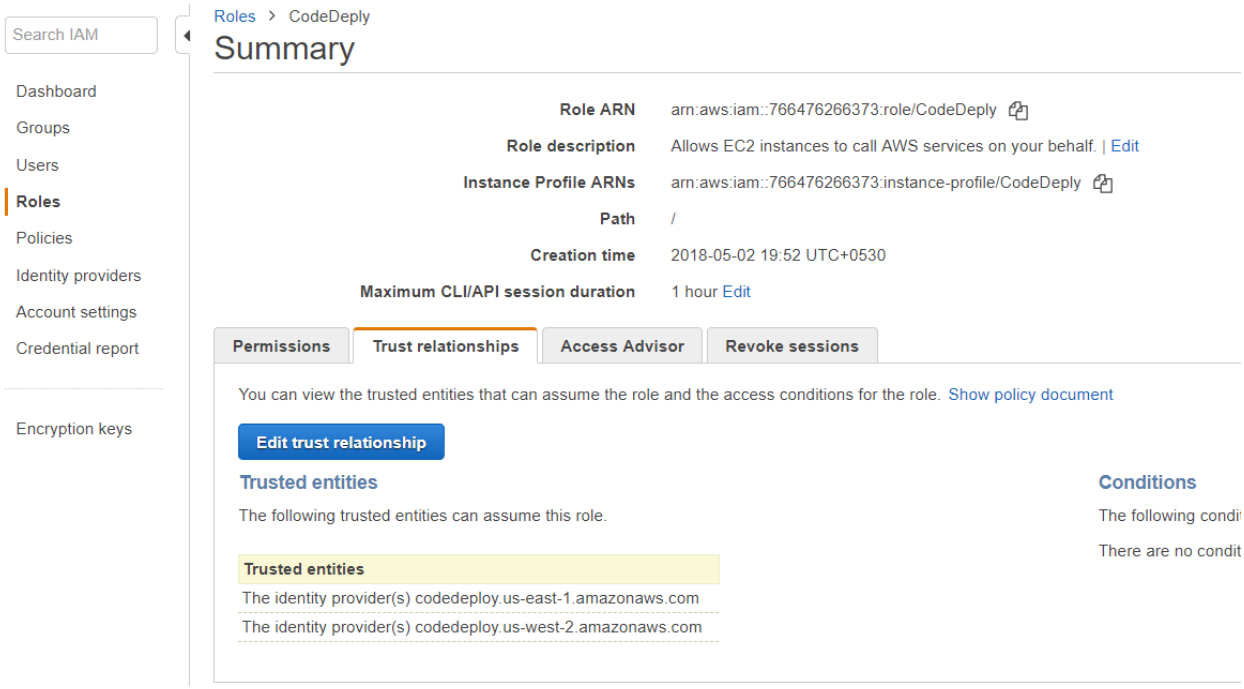
```
{
    "Version": "2012-10-17",
    "Statement": [
        {
            "Sid": "VisualEditor0",
            "Effect": "Allow",
            "Action": [
                "ec2:*"
            ],
            "Resource": "*"
        }
    ]
}
```

IAM Role for **"CodeDeploy"**



Then attach the above Codedeploy Policy to this IAM role

Add the below trust relationship on the CodeDeploy Role



Click on “Edit trust relationship”

```
{
   "Version": "2012-10-17",
   "Statement": [
      {
         "Sid": "",
         "Effect": "Allow",
         "Principal": {
            "Service": [
               "codedeploy.us-east-1.amazonaws.com",
               "codedeploy.us-west-2.amazonaws.com"
            ]
         },
         "Action": "sts:AssumeRole"
      }
   ]
}
```

Step 5: Create an CodeDeploy deployment code.

a.

CodeDeploy helps you to quickly deploy applications to Amazon EC2 instances or on-premises instances.

by creating a deployment that uses a sample application supplied by AWS CodeDeploy, or skip this wizard and
e a custom deployment with your own application.

Sample deployment

Recommended for new AWS CodeDeploy users.

Custom deployment

Recommended if you already have instances and an application to deploy.

Cancel | Skip Walkthrough

AWS CodeDeploy

Create application

Create an application and choose a deployment type. Specify the instances to deploy to. Specify the conditions

Application name* WEBAPP

Compute Platform* EC2/On-premises

Deployment group name* WEBAPPDG01

Deployment type

Choose the deployment to use to deploy your application. Learn more

In-place deployment

Updates the instances in the deployment group with the latest application revision. During
briefly taken offline for its update.

Blue/green deployment

Replaces the instances in the deployment group with new instances and deploys the latest
instances in the replacement environment are registered with a load balancer, instances fr
deregistered and can be terminated.

b.

Environment configuration

Environment configuration

Specify any combination of Auto Scaling groups, Amazon EC2 instances, and on-premises instances to add instances to this deployment group.

Auto Scaling groups | Amazon EC2 instances | On-premises instances

You can add up to three groups of tags for EC2 instances to this deployment group. Learn more
One tag group : Any instance identified by the tag group will be deployed to.
Multiple tag groups : Only instances identified by all the tag groups will be deployed to.

Tag group 1

	Key	Value	Instances	
1	Name	WS01	2	
2		WS01		

Add tag group

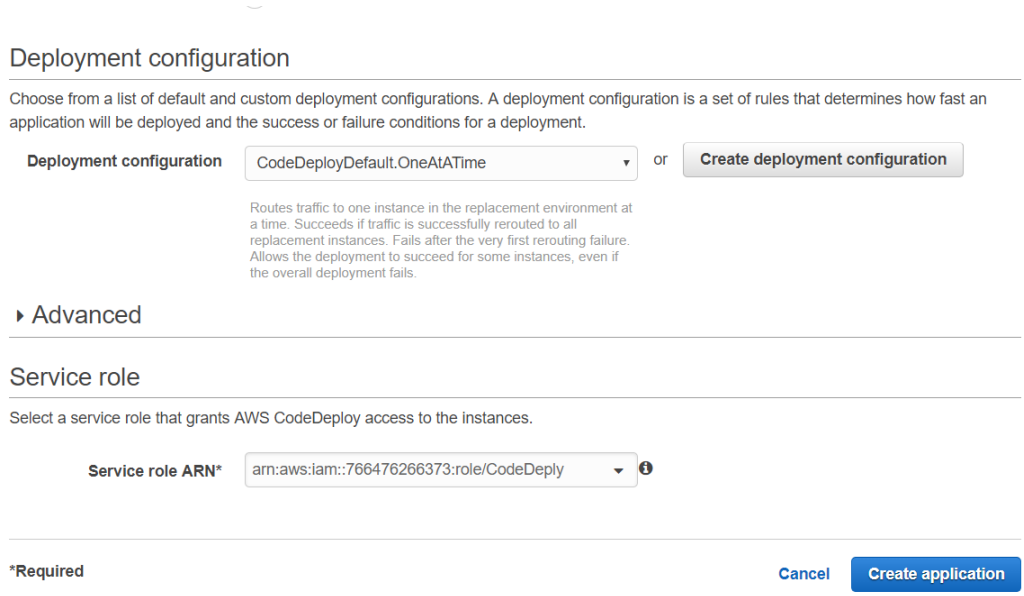
Matching instances

Right now these instances match the criteria you specified. There might be more or fewer instances that match your criteria when a deployment runs.

1 to 2 of 2 instances

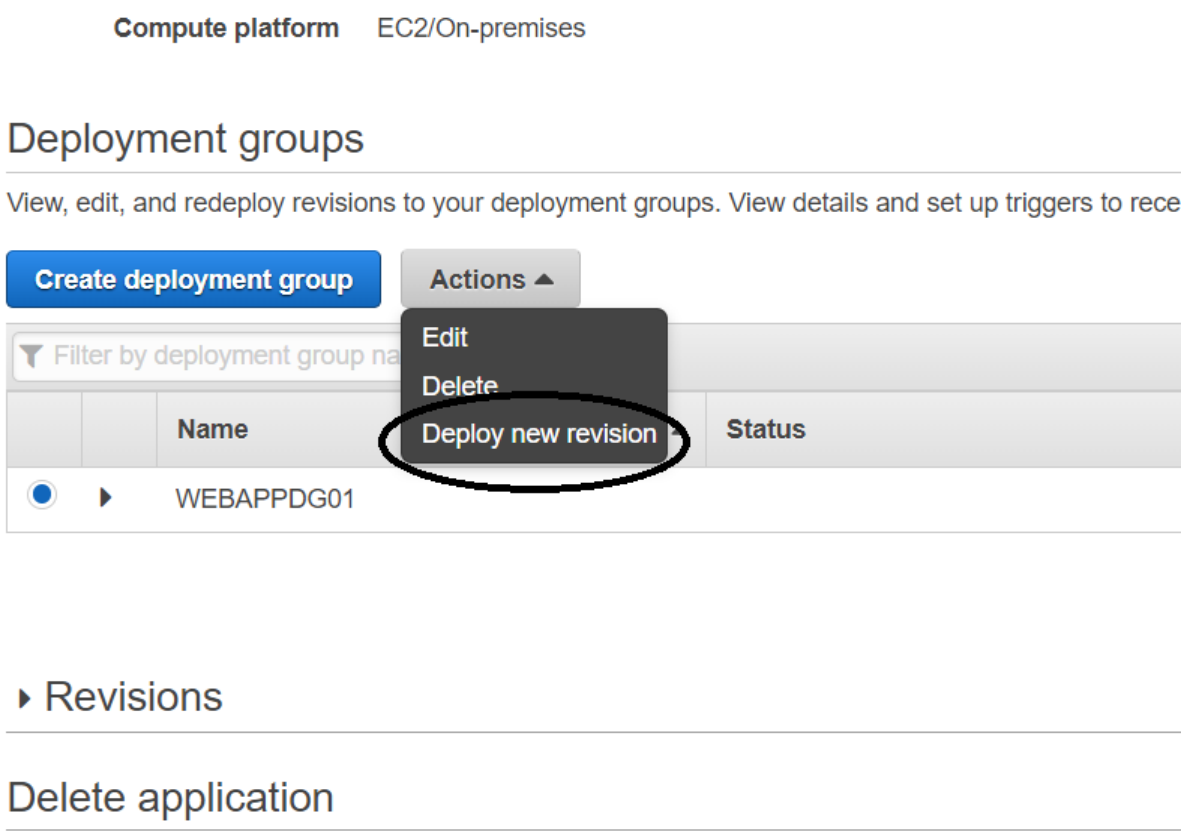
Instance ID	Status	Filter types	Association
i-064cb59c2f51d1e66	Running	Name:WS01	EC2
i-0c7b7ceaa6cbddf52	Running	Name:WS01	EC2

c.



Successfully Application Deployment Group has been created.

Step 6: Create an 'New Deploy" and point to the S3 location of the Sample-linux.zip file



Click on **"deploy new revision"**on the above screen

Deployment type In-place

Repository type* My application is stored in Amazon S3
My application is stored in GitHub

Revision location* https://s3.amazonaws.com/vishwa19820715/Sampl

Copy and paste the URL of your revision file location from Amazon S3.

Uploaded to Amazon S3 Sat, 19 May 2018 06:34:32 GMT

The date and time that your revision was uploaded to Amazon S3.

Version ID HfAbZBhxJ1Bx3piWt9v.QtheH.Rk7ARa

The version of the revision to be deployed.

File type* .zip **Supported file type**

The specified file has been detected as an accepted file type (.zip, .tar, .tar.gz).

Deployment description Add a brief description about the deployment. 100 character limit.

Copy the "S3 zip file link" to locate the zip file as above.

ApplicationStop lifecycle event failure

☑ Don't fail the deployment to an instance if this lifecycle event on the instance fails Learn more

Content options

Choose the action for AWS CodeDeploy to take during a deployment when a file on a target instance has the same name as a file in the application revision, for the same target location. Learn more

○ **Fail the deployment**

An error is reported and the deployment status is changed to Failed.

◉ **Overwrite the content**

The file in the application revision is copied to the target location on the instance, replacing the previous file.

○ **Retain the content**

The file in the application revision is not copied to the instance. The existing file is kept at the target location and treated as part of the new deployment.

Rollback configuration overrides

You can configure rollback options that apply only to this deployment. The rollback options you select here are used instead of those configured for the deployment group.

☐ Roll back when a deployment fails

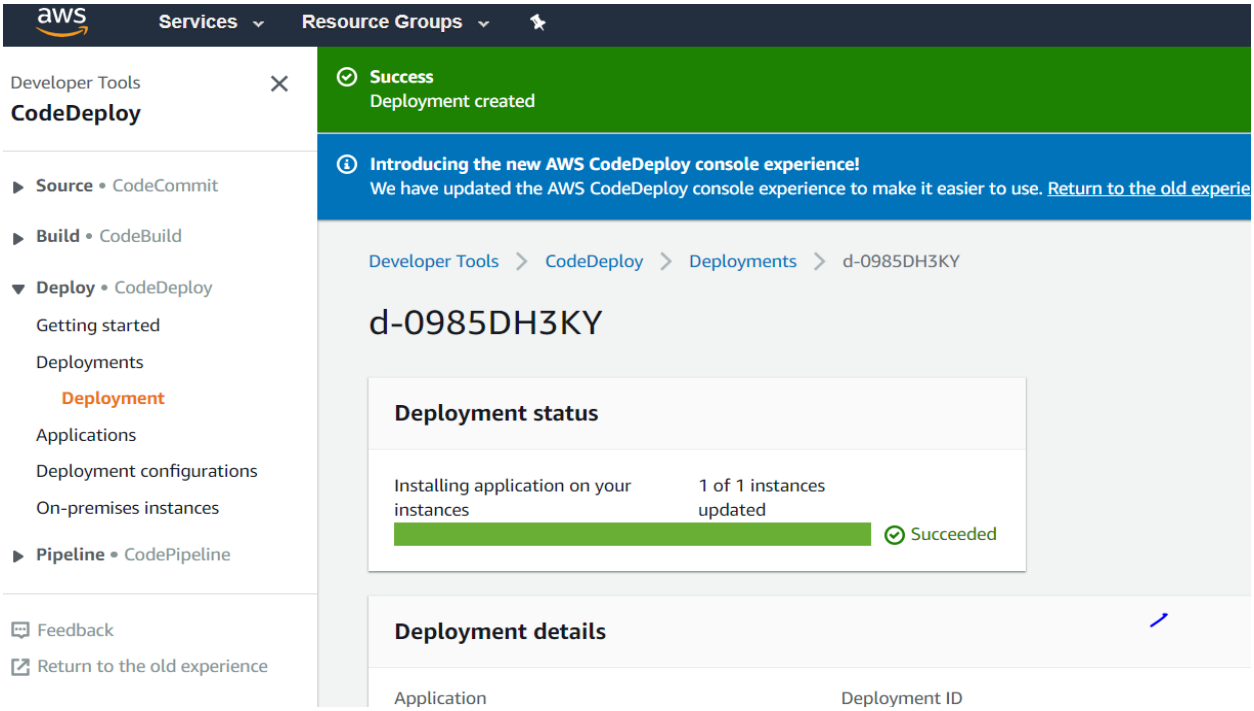
☐ Roll back when alarm thresholds are met

☑ Disable rollbacks

*Required

Cancel Deploy

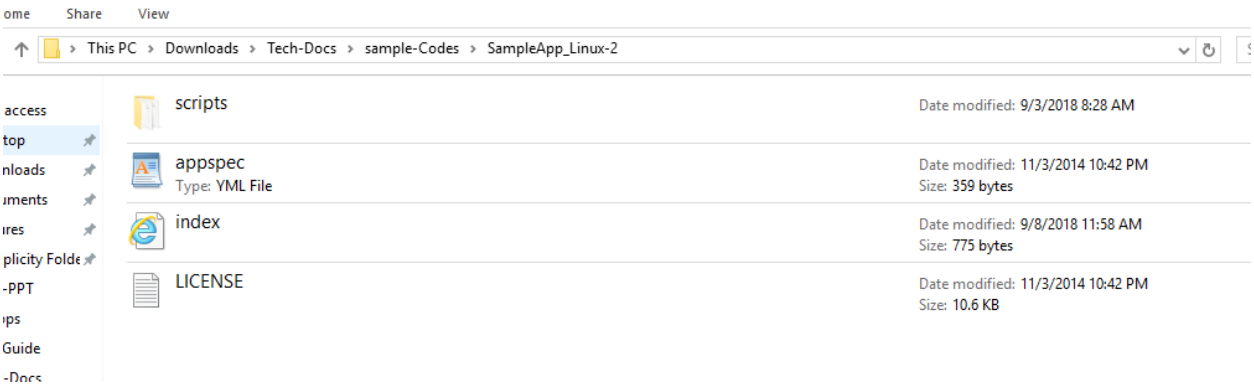
Select "Overwrite the content" if content found in the EC2 instance.



The Above screen would take upto 5 min as there are 2 instance in this example.

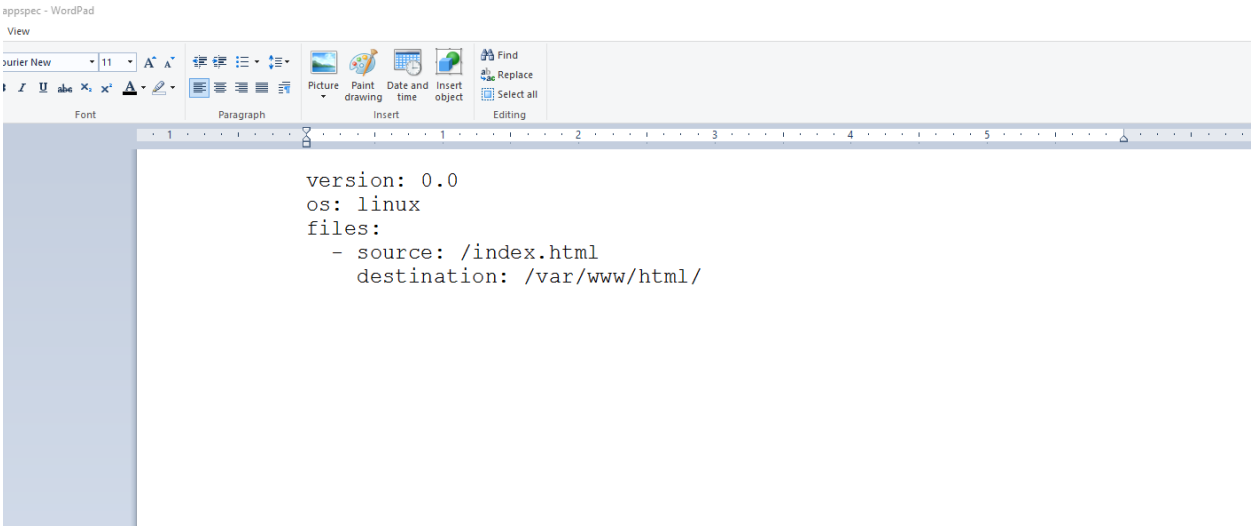
Step 7: Update the Deploy with new content

Download the Zip file on your laptop, uncompress it to a folder.



Remove the “Scripts” folder.

Edit the appspec file as below. Remove all other lines.



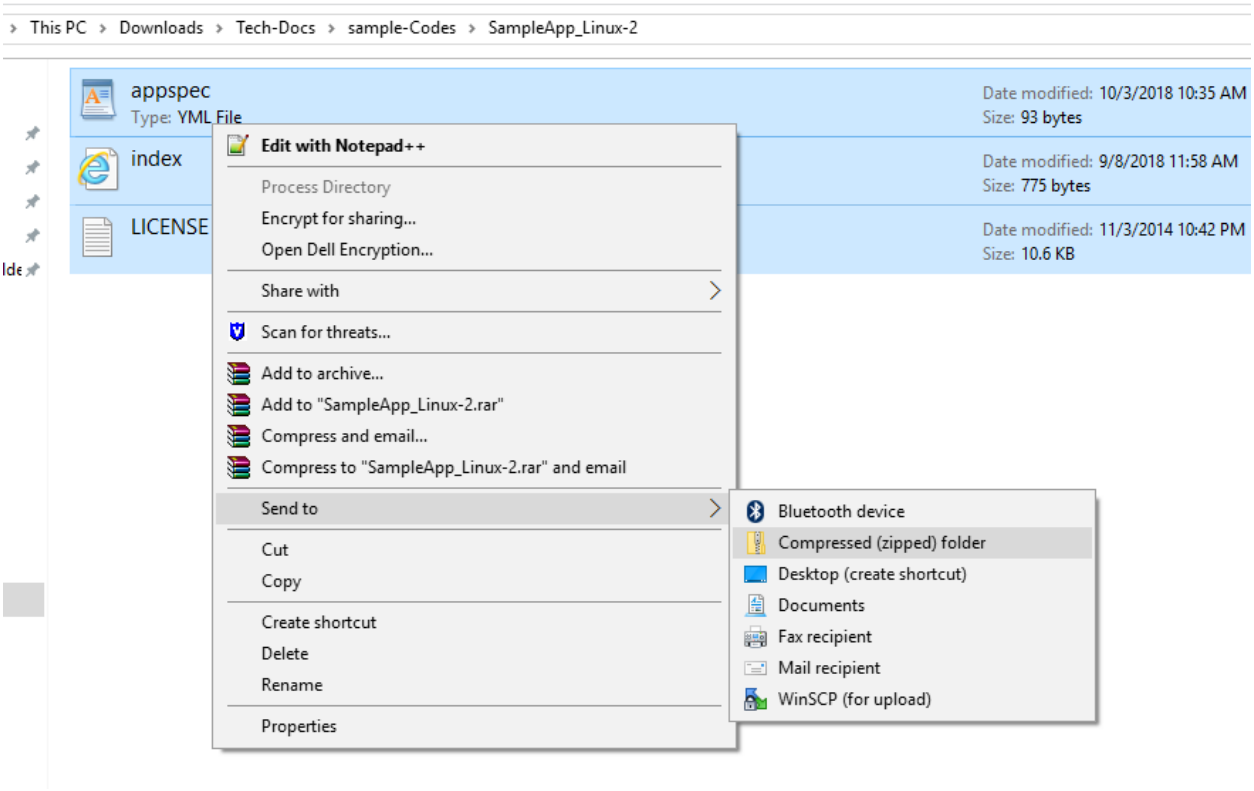
Now edit the “index.html” file. And add some content as below. And save it.

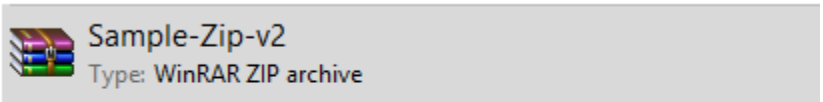
```
index.html
<!DOCTYPE html>
<html>
<head>
  <meta charset="utf-8">
  <title>Sample Deployment</title>
  <style>
    body {
      color: #ffffff;
      background-color: #0188cc;
      font-family: Arial, sans-serif;
      font-size: 14px;
    }

    h1 {
      font-size: 500%;
      font-weight: normal;
      margin-bottom: 0;
    }

    h2 {
      font-size: 200%;
      font-weight: normal;
      margin-bottom: 0;
    }
  </style>
</head>
<body>
  <div align="center">
    <h1>Congratulations Batch, HTTP service is UP</h1>
    <h2>This application was deployed using AWS CodeDeploy. But was customized by Vishwa</
    <p>For next steps, read the <a href="http://aws.amazon.com/documentation/codedeploy">A
    Documentation</a>.</p>
  </div>
</body>
</html>
```

Now zip the file as shown. Select only the FILES and zip and NOT the folder itself, as CODEDEPLOY would be looking for “appsepc” file in the zip parent folder.





Now upload this Sample-zip-v2 to S3, and make it public.

Update the same Deployment Group with “new Revision” with new S3 link.